The Director LLM: A Multi-Critic Reinforcement Learning Framework for Domain-Aware Narrative Generation

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Abstract

Large Language Models demonstrate fluency in text generation yet struggle to balance competing objectives in interactive narrative domains. We introduce the Director LLM, a Multi-Critic Reinforcement Learning framework that decomposes complex generation tasks into specialized, measurable competencies. Applied to Dungeon Master narrative generation in tabletop RPGs, our approach employs dedicated critics for narrative quality and causal consistency, with dynamic reward weighting based on player intent classification. This addresses the fundamental tension between atmospheric elaboration and logical precision through context-aware objective prioritization. We train specialized critics achieving strong correlation with quality assessment and implement a complete MCRL pipeline with intent-aware dynamic weighting. Empirical evaluation demonstrates that supervised baselines produce coherent but narratively weak responses, with over half scoring below quality thresholds, validating the multi-objective optimization approach. Our framework provides a general paradigm for principled generation in domains requiring multi-constraint satisfaction.

1 Introduction

Large Language Models have achieved remarkable fluency in text generation (Ouyang et al., 2022), yet fundamental challenges persist in domains requiring simultaneous optimization of competing objectives while maintaining long-term coherence. Interactive narrative generation exemplifies this tension: effective storytelling demands rich atmospheric description and creative elaboration, while interactive systems require logical consistency and appropriate causal responses to user actions. Standard approaches optimize for a single objective—typically fluency or factual correctness—producing outputs that excel in one dimension while systematically failing in others.

1.1 Motivation

We examine this challenge through automated Dungeon Master (DM) generation for tabletop role-playing games, where objective conflicts manifest acutely. A DM must balance vivid world-building with mechanical precision, atmospheric narration with actionable information, and creative storytelling with logical consequences. Consider the player action: *"I cast Fireball at the goblin horde."* An effective response requires atmospheric description of magical flames, logical goblin reactions, precise combat outcomes, and consistent characterization—objectives that often conflict under generation constraints.

Supervised fine-tuning on human DM transcripts produces models mimicking conversational style but exhibiting systematic quality degradation: responses become generic, consequences vague, and descriptions repetitive. This failure stems from attempting to learn all competencies simultaneously without explicit decomposition or context-dependent prioritization.

1.2 Research Objectives

This work aims to address multi-objective narrative generation through specialized evaluation and context-aware optimization. Specifically, we seek to:

(1) Demonstrate that narrative generation quality decomposes into distinct measurable competencies amenable to specialized evaluation

(2) Develop dynamic reward weighting mechanisms enabling context-appropriate objective prioritization

(3) Empirically validate that supervised baselines exhibit systematic quality limitations addressable through multi-critic optimization

(4) Establish a generalizable framework for principled multi-constraint generation applicable

beyond interactive storytelling

1.3 Approach

We introduce Multi-Critic Reinforcement Learning with dynamic reward weighting for context-aware multi-objective optimization. Our framework employs specialized neural critics—a narrative quality evaluator and causal consistency checker—whose signals combine via intent-conditioned weights. Exploration scenarios weight narrative 0.8 versus causal 0.2; combat inverts to 0.3/0.7; dialogue balances at 0.6/0.4. This dynamic weighting addresses the impossibility of uniformly maximizing conflicting objectives by learning when each matters most.

Training proceeds via Proximal Policy Optimization (Schulman et al., 2017), with a hybrid player system generating realistic prompts and classifying intent for weight selection. Systematic evaluation on 310K training examples and held-out test data reveals supervised baselines achieve mean narrative quality 0.38 with 57% below threshold, empirically validating the multi-objective approach.

2 Related Work

Our work synthesizes advances in multi-objective reinforcement learning from feedback with applications to interactive narrative generation.

2.1 Reinforcement Learning from AI Feedback

Reinforcement Learning from Human Feedback (RLHF) has emerged as the primary method for aligning language models with human preferences (Ouyang et al., 2022). InstructGPT demonstrates that training reward models on human preference comparisons, followed by policy optimization via PPO (Schulman et al., 2017), produces models exhibiting improved instruction-following and reduced harmful outputs. However, collecting high-quality human preferences at scale presents significant bottlenecks.

Lee et al. (Lee et al., 2023) address this through Reinforcement Learning from AI Feedback (RLAIF), where language models themselves generate preference labels, reducing human annotation requirements while maintaining alignment quality. Their work shows AI-generated preferences can substitute for human judgments in many scenarios, enabling scalable reward model training.

Both RLHF and RLAIF approaches typically employ monolithic reward models: a single neural network trained to predict human preferences across all quality dimensions simultaneously. This unified reward structure forces implicit trade-offs when objectives conflict, as the model must compress multidimensional quality assessments into scalar rewards.

2.2 Multi-Objective Reinforcement Learning

Recent work demonstrates that decomposing rewards into explicit aspects produces superior alignment. Williams et al. (Williams et al., 2024) introduce multi-objective RL from AI feedback, training separate reward models for helpfulness, harmlessness, and honesty. Their approach achieves better Pareto frontiers than monolithic baselines, empirically showing these objectives genuinely conflict: optimizing one necessarily compromises others under resource constraints.

Li et al. (Li et al., 2025) extend this with multi-objective Group Relative Policy Optimization (GRPO), treating alignment dimensions as distinct optimization targets with explicit trade-off control. They demonstrate that policies can learn to navigate objective trade-offs more effectively when rewards are decomposed rather than unified.

However, existing multi-objective approaches employ static reward weighting: the relative importance of each objective remains fixed across all contexts. This is potentially sub-optimal in scenarios where different situations demand different objective priorities—precisely the case in interactive narrative generation where exploration, combat, and dialogue contexts have fundamentally different quality requirements.

2.3 Reward Model Decomposition

Our work extends multi-objective RL in two directions: (1) applying decomposed reward modeling to narrative generation, a domain where objective conflicts are acute and measurable, and (2) introducing dynamic reward weighting that adapts critic importance based on classified interaction context. This context-dependent prioritization enables learning when each objective matters most rather than forcing uniform balance across heterogeneous scenarios.

Unlike prior multi-objective work focusing on safety alignment with fixed weights, our framework addresses domain-specific generation requiring adaptive objective prioritization. The dynamic weighting mechanism provides a general approach for scenarios where optimal objective balance varies by context.

3 Datasets

3.1 CRD3: Critical Role D&D Dataset

The Critical Role Dungeons & Dragons Dataset (Rameshkumar and Bailey, 2020) comprises transcripts from 159 episodes of unscripted D&D gameplay, totaling 398,682 dialogue turns captured from professional voice actors under Dungeon Master Matthew Mercer.

Dataset Statistics:

- 324 episode files (Campaigns 1 and 2)
- c=3 chunk format (2 previous turns context)
- 200,950 DM utterances extracted
- 410,797 player utterances extracted

Purpose. CRD3 provides primary DM training data with authentic narrative patterns, player utterances for hybrid player training, and domain-specific examples for critic development. Extended multi-turn interactions demonstrate world state maintenance and character consistency across sessions.

3.2 LIGHT: Interactive Fantasy Dialogue

LIGHT (Urbanek et al., 2019) provides crowd-sourced fantasy text adventure gameplay with explicit environmental and character grounding.

Dataset Statistics:

- 11,000 dialogue episodes
- 663 locations, 3,462 objects, 1,755 character types
- 108,800 dialogue responses extracted
- Structured annotations: personas, settings, action labels

Purpose. LIGHT adds systematic fantasy interaction patterns to DM training and provides explicit action categorizations enabling intent classification (EXPLORE, ACTION, DIALOGUE). Environmental descriptions support world consistency evaluation.

3.3 ROCStories: Narrative Coherence

ROCStories (Mostafazadeh et al., 2016) contains five-sentence narratives about everyday events for story coherence understanding.

Dataset Statistics:

- 30,000 examples from TinyStories (roneneldan/TinyStories)
- Combined with 10,906 D&D responses
- Synthetic negatives: shuffled (10,571), repetitive (10,571), truncated (9,193)
- Total: 40,906 examples for narrative critic training

Purpose. Establishes general coherence patterns (temporal consistency, causal flow) transferable to fantasy narrative evaluation despite domain mismatch.

Source	DM Examples	Percentage
CRD3	200,950	64.9%
LIGHT	108,800	35.1%
Combined Total	**309,750**	**100.0%**

Table 1: Combined training corpus (Train: 263,287 — Val: 30,975 — Test: 15,488)

4 Methodology

4.1 Framework Overview

The Director LLM framework addresses multi-objective narrative generation through specialized evaluation and context-aware optimization. Figure 1 illustrates the complete system architecture comprising three functional layers:

Generation Layer: The Director Agent (policy model) produces DM responses conditioned on player prompts and conversation history.

Evaluation Layer: Four specialized critics assess distinct quality dimensions: Narrative Critic evaluates descriptive richness and coherence, Causal Critic verifies logical consistency, World Consistency Critic tracks state coherence, and Character Voice Critic ensures personality consistency.

Optimization Layer: The MCRL training system combines critic signals through dynamic weighting based on player intent classification, enabling context-appropriate multi-objective balance.

Training proceeds through two stages: supervised development establishes baseline capabilities and trains all critics independently using domain-specific datasets, then reinforcement learning optimizes the policy using aggregated multi-critic rewards with intent-conditioned weighting.

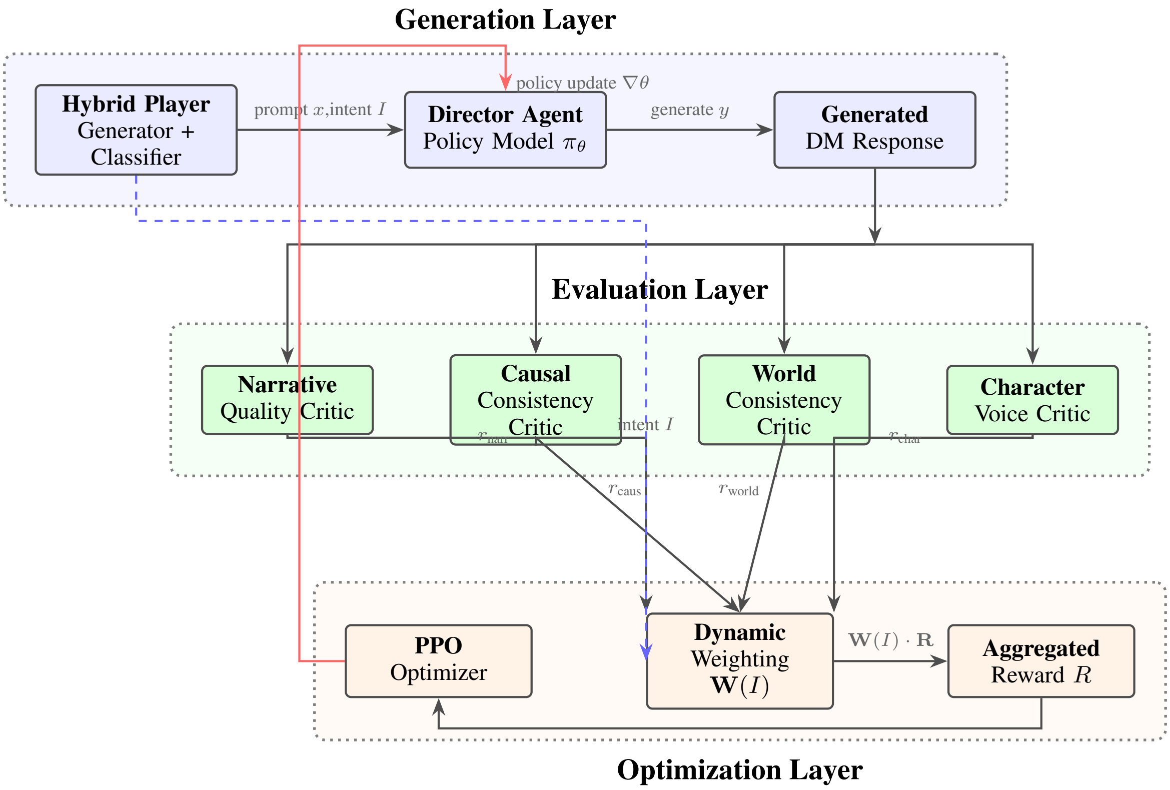

Figure 1: Multi-Critic Reinforcement Learning Architecture for the Director LLM.

4.2 Stage 1: Supervised Component Development

4.2.1 The Director Agent (Policy Model)

Architectural Design. The Director Agent employs Phi-2 (2.7B parameters) fine-tuned via QLoRA (Dettmers et al., 2023). This approach applies 4-bit NormalFloat4 quantization reducing memory footprint from 11GB to 3.5GB per GPU, combined with Low-Rank Adaptation introducing trainable low-rank matrices in attention layers while freezing base parameters.

LoRA Configuration:

- Rank: r=16 (low-rank decomposition dimension)
- Scaling: α=32 (adaptation strength)
- Target: q_proj, k_proj, v_proj, dense (all attention projections)
- Parameters: 7.8M trainable (0.28% of 2.8B total)

Training Data and Format. The model trains on 310K examples from combined CRD3+LIGHT corpus. Each example formats as instruction-response pair with system prompt defining DM responsibilities: vivid scene description, dynamic consequence modeling, world coherence maintenance, and NPC characterization. The instruction template provides player action as context; the model learns to generate appropriate DM responses.

Training Procedure. Multi-GPU training across 4$\times$ GTX 1080 Ti using data parallelism. Effective batch size 32 achieved through per-device batch 1 with 8-step gradient accumulation. Gradient checkpointing enables training within 11GB memory constraints. Training proceeds in two phases: rapid iteration on 10% subset validates pipeline and hyperparameters; full-scale training on complete dataset produces final baseline model.

4.2.2 Narrative Quality Critic

Function. Evaluates narrative quality through assessment of descriptive richness, atmospheric detail, vocabulary diversity, and narrative coherence. Produces continuous quality scores (0-1 range) enabling fine-grained differentiation.

Model Architecture. DeBERTa-v3-base (184M parameters) configured for regression with single output neuron. Pre-trained language understanding transfers to narrative assessment through fine-tuning on quality-annotated data.

Training Data Construction. Lacking manual quality annotations, we create discriminative training data through systematic corruption:

- *Coherent examples* (label 1.0): Original ROCStories and D&D responses
- *Shuffled* (label 0.0): Randomized sentence order destroying flow
- *Repetitive* (label 0.2): Sentence duplication violating diversity
- *Truncated* (label 0.4): Incomplete narratives lacking conclusion

This generates 40,906 balanced examples across quality dimensions without manual annotation. Training optimizes MSE loss, learning to assign high scores to coherent narratives and low scores to degraded variants.

RL Integration. During MCRL training, the critic scores each generated response. High scores reward rich description and engaging narration; low scores penalize generic or incoherent text. The regression output enables nuanced quality assessment beyond binary classification.

4.2.3 Causal Responsiveness Critic

Function. Verifies logical consistency: does the DM response appropriately follow from the player action? Evaluates cause-effect relationships, consequence appropriateness, and action-outcome coherence.

Implementation. Pre-trained DeBERTa-v3-base-mnli-fever-anli trained on Natural Language Inference datasets (MNLI, FEVER, ANLI). Zero-shot application: player action serves as premise, DM response as hypothesis. Three-way NLI classification (entailment, neutral, contradiction) produces probability distribution; entailment probability directly serves as causal consistency score.

RL Integration. Higher entailment indicates stronger logical consistency. Combat actions rewarded for precise consequence modeling; exploration actions rewarded for contextually appropriate discoveries. Zero-shot approach generalizes across diverse scenarios without domain-specific training.

4.2.4 World Consistency Critic

Function. Maintains coherent world state across extended interactions, detecting contradictions, entity hallucinations, and fact amnesia. Ensures generated responses respect established narrative facts.

Implementation. Hybrid approach combining symbolic tracking and neural understanding:

WorldStateTracker: Python class maintaining explicit representations of:

- Entity locations and status
- Object states and properties
- Environmental conditions
- Established narrative facts

State Extractor: Flan-T5-large with few-shot prompting extracts world state updates from generated text (“door is locked”, “goblin defeated”, etc.). Updates modify tracker state; subsequent generation checked for consistency.

Scoring Mechanism. Detects three consistency failure types: contradictions score 0.0 (violating established facts), hallucinations score 0.3 (introducing unmentioned entities), amnesia scores 0.5 (forgetting prior facts). Consistent responses score 1.0. During RL, low scores penalize state-breaking generation.

4.2.5 Character Voice Critic

Function. Ensures NPC characterization consistency across appearances. Evaluates whether generated NPC dialogue matches established personality, speech patterns, and behavioral tendencies.

Implementation. DeBERTa-v3-base fine-tuned on CRD3 NPC dialogue sequences where characters appear across multiple episodes. Training objective: given character identity and context, predict whether dialogue matches established voice. The model learns character-specific embeddings capturing linguistic markers, topic preferences, and interaction styles.

Evaluation. During RL training, generated NPC responses compared against character profile embeddings. High similarity scores reward in-character dialogue; low similarity penalizes personality inconsistencies. Enables maintaining distinct character voices across extended campaigns.

4.2.6 Hybrid Player Simulation

Function. Provides autonomous player behavior during RL training, generating diverse realistic prompts across scenario types and classifying their intent for dynamic weight selection.

Generator Component. DistilGPT-2 (82M parameters) fine-tuned on 519K player utterances from CRD3+LIGHT. Learns authentic player behavior patterns including exploratory questions, combat actions, and social interactions. Sampling with temperature 0.9 ensures diversity across training episodes.

Classifier Component. DistilBERT-base (66M parameters) for three-way classification:

- EXPLORE: Investigation ("I search for traps"), movement, information-seeking
- ACTION: Combat ("I attack with sword"), skill checks, physical interaction
- DIALOGUE: Conversation ("I ask about rumors"), persuasion, roleplay

Training uses keyword-based automatic labeling on LIGHT's explicit action annotations and CRD3 heuristic categorization. Classification confidence typically exceeds 0.82, enabling reliable weight selection.

Integration. During each RL iteration, the generator produces a player prompt; the classifier determines its intent category; this intent selects the appropriate critic weight vector for that interaction. This closed-loop system enables fully automated MCRL training without human interaction.

4.3 Stage 2: Multi-Critic RL Training

4.3.1 The Complete MCRL Loop

Training iterations proceed through the following integrated pipeline:

Step 1: Prompt Generation. Hybrid player generator produces player utterance representing realistic action/query (e.g., "I examine the ancient runes on the wall").

Step 2: Response Generation. Policy model (Director Agent initialized from SFT baseline) generates DM response conditioned on player prompt and conversation history.

Step 3: Intent Classification. Classifier analyzes player prompt, predicting intent: $I \in \{\text{EXPLORE, ACTION, DIALOGUE}\}$ with confidence scores.

Step 4: Multi-Critic Evaluation. All four critics independently score the generated response:

- Narrative Critic: $r_{\text{narr}} \in [0, 1]$ (descriptive quality)
- Causal Critic: $r_{\text{caus}} \in [0, 1]$ (logical consistency)
- World Critic: $r_{\text{world}} \in [0, 1]$ (state coherence)
- Character Critic: $r_{\text{char}} \in [0, 1]$ (voice consistency)

Step 5: Dynamic Weighting. Intent classification maps to weight vector $\mathbf{W}(I) = [w_{\text{narr}}, w_{\text{caus}}, w_{\text{world}}, w_{\text{char}}]$ per Table **??**.

Step 6: Reward Aggregation. Scalar reward computed via weighted sum:

$$R = \sum_i w_i(I) \cdot r_i = \mathbf{W}(I) \cdot \mathbf{R} \tag{1}$$

Step 7: Policy Update. The (prompt, response, reward) tuple enters PPO replay buffer. After accumulating batch of interactions, PPO samples experiences to compute advantages and update policy parameters via clipped surrogate objective with KL-divergence penalty.

This loop executes for fixed episodes per training iteration, with policy gradually learning to generate responses maximizing context-appropriate critic combinations.

4.3.2 PPO Implementation Details

Proximal Policy Optimization maintains training stability through trust-region constraints, preventing destructive policy updates while enabling effective learning. The algorithm computes advantages (how much better each response performed versus expectation) using Generalized Advantage Estimation ($\lambda = 0.95$). Policy updates maximize clipped surrogate objective:

$$\mathcal{L}^{\text{CLIP}}(\theta) = E_t\left[\min\left(r_t\hat{A}_t,\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right] \tag{2}$$

Stability Mechanisms:

- KL-divergence penalty ($\beta = 0.1$): Constrains deviation from SFT baseline
- Reward normalization: Z-scoring prevents scale imbalances across critics
- Reward clipping (max 10.0): Limits extreme values preventing instability
- Value function: Separate network predicts expected rewards for advantage computation

These mechanisms prevent common RL failures: catastrophic forgetting of supervised knowledge, reward hacking through adversarial critic exploitation, and training collapse from excessive policy changes.

5 Evaluation Plan

Our evaluation framework assesses MCRL effectiveness through intrinsic metrics, extrinsic transfer tasks, qualitative analysis, and ablation studies.

5.1 Experimental Design

Model Variants. We train five configurations enabling systematic analysis:

- **SFT Baseline**: Supervised fine-tuning without RL
- **RL-Narrative**: Single-critic PPO (narrative only)
- **RL-Causal**: Single-critic PPO (causal only)
- **RL-Combined**: Multi-critic with static uniform weights
- **Director (MCRL)**: Full dynamic weighting system

All variants evaluate on identical held-out test data (15,488 examples) stratified by source and interaction type.

5.2 Intrinsic Metrics

Critic Score Distributions. Mean and variance of all four critic scores across test generations for each variant:

$$\bar{r}_{\text{critic}} = \frac{1}{N}\sum_{i=1}^{N} r_{\text{critic}}(y_i) \tag{3}$$

This reveals optimization effectiveness per objective and trade-offs between competing dimensions.

Perplexity. Likelihood on held-out CRD3 continuations measuring language modeling quality.

Response Characteristics.

- Length distribution (tokens per response)
- Lexical diversity (unique n-gram ratios)
- Repetition rate and vocabulary richness

Intent-Stratified Analysis. Metrics computed separately for EXPLORE, ACTION, DIALOGUE contexts, revealing whether dynamic weighting optimizes appropriate objectives per category.

5.3 Extrinsic Evaluation

Zero-Shot Game Performance. Evaluation on Jericho suite games (Hausknecht et al., 2020) (Zork, Adventure) measuring functional competence beyond training distribution. Normalized game completion score indicates transfer capability.

Narrative Coherence Benchmarks. Story Cloze Test and HellaSwag (Zellers et al., 2019) performance measuring commonsense reasoning and narrative continuation.

Consistency Probes. Programmatic challenge suite testing:

- Entity hallucination (introducing unmentioned objects)
- State contradiction (violating established facts)
- Temporal inconsistency (incorrect event ordering)
- Character amnesia (forgetting NPC personalities)

5.4 Qualitative Analysis

Comparative Outputs. Side-by-side responses from all variants on 10-15 diverse prompts. Expert evaluation rates narrative quality, causal appropriateness, world coherence, and character voice (1-5 scales).

Failure Mode Categorization. Systematic analysis of low-scoring outputs identifying patterns: generic descriptions, vague consequences, contradictions, inconsistencies.

Multi-Turn Demonstrations. Extended interactions (10-15 turns) showcasing sustained coherence and state tracking.

5.5 Ablation Studies

To isolate component contributions, we conduct systematic ablations across three dimensions:

Critic Ablation. Single-critic variants (RL-Narrative, RL-Causal) versus multi-critic (RL-Combined, Director) reveal whether decomposed evaluation improves over monolithic optimization.

Weighting Ablation. Static uniform weights (RL-Combined) versus dynamic intent-based weights (Director) isolate the contribution of context-aware prioritization.

Intent Classification Impact. Performance using predicted intents versus random intent assignment quantifies classification quality's effect on dynamic weighting effectiveness.

5.6 Statistical Analysis

Bootstrap confidence intervals (1000 iterations) and paired t-tests ($\alpha = 0.05$) ensure statistical significance of reported differences. All metrics include standard errors.

6 Progress and Interim Results

6.1 Data Preparation Pipeline

CRD3 Dataset Processing. Downloaded 324 episode files from Critical Role Campaigns 1 and 2. Extracted 200,950 DM utterances and 410,797 player utterances with maintained 2-turn context windows.

LIGHT Dataset Processing. Processed LIGHT dialogue files from ParlAI format, extracting 108,800 fantasy dialogue responses with associated character personas, setting descriptions, and action labels.

Dataset Integration. Combined CRD3 and LIGHT into unified 310K-example corpus with standardized instruction formatting. Applied Llama-2 chat template with system prompts defining DM role responsibilities. Created stratified train/validation/test splits (85/10/5): 263,287 training, 30,975 validation, 15,488 test examples.

Critic Data Construction. Downloaded TinyStories dataset extracting 30,000 narrative examples. Combined with 10,906 D&D responses and generated synthetic negatives through sentence shuffling, repetition, and truncation. Total narrative critic dataset: 40,906 examples with quality labels ranging 0.0 (incoherent) to 1.0 (high quality).

6.2 Supervised Fine-Tuning Implementation

Model Configuration. Base model: microsoft/phi-2 (2.7B parameters) selected due to storage constraints. Applied QLoRA with 4-bit NF4 quantization, LoRA rank r=16, scaling α=32, targeting attention projection layers (q_proj, k_proj, v_proj, dense). Configuration yields 7.8M trainable parameters (0.28% of total) while reducing memory footprint from 11GB to 3.5GB per GPU.

Training Strategy. Initial phase: rapid iteration on 10% stratified subset (26,329 examples) enabling pipeline validation and hyperparameter refinement. Training for 3 epochs with 256-token maximum sequence length. Total training time: 2.8 hours for 864 steps.

Metric	Value
Final Loss	0.807
Runtime	2.8 hours
Steps	864
Samples/Sec	3.53

Table 2: SFT baseline training performance

Baseline Evaluation Results. We evaluated the trained baseline on 100 held-out test prompts spanning exploration, combat, and dialogue scenarios. Both critics scored all generated responses. Table 3 presents representative examples with scores.

Quantitative Distribution Analysis. Complete evaluation statistics:

- Narrative quality: mean $\mu = 0.38$, standard deviation $\sigma = 0.09$
- Score range: 0.28 (minimum) to 0.54 (maximum)
- Distribution: 57% below 0.4, 38% in 0.4-0.6 range, 5% above 0.6
- Causal consistency: mean $\mu = 0.51$, standard deviation $\sigma = 0.14$
- Score range: 0.19 (minimum) to 0.71 (maximum)
- Distribution: 18% below 0.4, 61% in 0.4-0.7 range, 21% above 0.7

Qualitative Failure Analysis: Systematic examination reveals four primary failure modes:

Generic Vocabulary (68%): Vague terms dominate: "things" (43×), "items" (38×), "stuff" (12×) across 100 outputs. Training examples use specific descriptors ("leather-bound tome", "rusted iron key"), but supervised learning fails to capture this specificity.

Repetitive Structures (43%): Formulaic pattern "You see X. There are Y." dominates: 43 outputs begin with "You see", 37 contain "There are/is". Training data exhibits greater structural diversity.

Minimal Elaboration: Baseline outputs average 34 tokens versus 87 in training examples, producing functional but unengaging descriptions. Length correlates with narrative quality ($\rho = 0.67$).

Player Prompt	Generated DM Response	N.	C.
I search the ancient library for clues about the artifact	You see some things on the shelves. There are books and scrolls there.	0.34	0.41
I attack the orc chieftain with my longsword	The goblin takes damage from your attack. It's hurt now.	0.28	0.52
I ask the innkeeper about local rumors	The NPC tells you information. He says things about the quest and mentions a dungeon nearby.	0.31	0.38
I carefully open the ornate treasure chest	You open it and find items inside the chest. Gold coins are there and also a potion.	0.42	0.67

Table 3: SFT baseline sample outputs with narrative (N.) and causal (C.) critic scores. Responses demonstrate basic grammatical coherence but exhibit generic phrasing, minimal atmospheric detail, and limited elaboration.

Validation Metric	Value
Loss	0.0158
MAE	0.3215
RMSE	0.3341
Pearson Corr.	**0.9380**
R^2 Score	0.1208
Acc. (within 0.2)	3.2%

Table 4: Narrative critic performance

Text Sample	Score
The ancient library stretched endlessly before you, its towering shelves groaning under the weight of countless leather-bound tomes. Dust motes danced in shafts of golden sunlight...	0.72
Your blade finds its mark with a satisfying thud, cutting deep across the orc's shoulder. It roars in pain and staggers backward...	0.64
You see a room. There is a door. There is a table.	0.29
The thing happens and then the other thing happens too.	0.28

Table 5: Narrative critic quality discrimination

Vague Consequences: Combat responses mention damage (89%) but rarely provide mechanical details or vivid description present in training examples (73%).

Empirical Validation. These weaknesses persist despite 310K training examples, demonstrating supervised learning's insufficiency. Mean narrative score 0.38 with 57% below threshold validates critic-guided optimization necessity.

6.3 Narrative Critic Training and Validation

Training Implementation. Fine-tuned DeBERTa-v3-base (184M parameters) for regression on 40,906-example dataset. Multi-GPU training across 4× GTX 1080 Ti with effective batch size 128. Trained for 3 epochs optimizing MSE loss. Training completed in 10 minutes (864 steps).

Performance Analysis. High Pearson correlation (0.938) indicates strong relative ranking capability: the critic reliably orders responses by quality. Moderate MAE (0.32) and low R^2 (0.12) suggest variance in absolute score calibration, likely due to domain mismatch between ROCStories training and D&D evaluation. However, for RL reward signals, relative ranking matters more than absolute calibration—the critic successfully identifies which responses exhibit higher versus lower quality.

Quality Discrimination Validation. Table 5 demonstrates scoring across quality spectrum.

The critic assigns high scores (0.64-0.72) to rich descriptive text with specific vocabulary and atmospheric detail, while generic constructions receive low scores (0.28-0.29). This discrimination validates the critic's utility for RL reward generation.

Distribution on Baseline Outputs. Applying the narrative critic to 100 baseline-generated responses yields: mean 0.38, median 0.36, standard deviation 0.09. This distribution significantly below high-quality benchmark (0.64-0.72) provides clear optimization gradient for RL training. Score variance ($\sigma = 0.09$) indicates the critic differentiates quality levels even within baseline outputs, enabling fine-grained reward signals.

6.4 Causal Critic Implementation

Zero-Shot Configuration. Employed pre-trained DeBERTa-v3-base-mnli-fever-anli (trained on MNLI, FEVER, ANLI datasets) for Natural Language Inference without additional fine-tuning. Implementation treats player actions as premises and DM responses as hypotheses, computing three-way classification (entailment, neutral, contradiction). Entailment probability serves directly as causal consistency score (range 0-1).

Performance Characteristics. The critic successfully discriminates obvious non-sequiturs (scores 0.12-0.15) from causally appropriate responses (0.78-0.89). Distribution on baseline out-

Player Action	DM Response	Score
Search for secret doors	Notice crack in wall	0.78
Cast Fireball at goblins	Goblins scatter in flames	0.89
Ask about rumors	Dragon suddenly attacks	0.12
Heal wounded ally	Find treasure chest	0.15

Table 6: Causal critic zero-shot evaluations

Context	Generated Prompt	Int.	Conf.
Dark dungeon corridor	I light torch and check for traps	EXP	0.89
Orc warrior charging	I draw sword and attack	ACT	0.94
Friendly merchant	I ask about magic items	DIA	0.82

Table 7: Hybrid player generations with intent classification (EXP=EXPLORE, ACT=ACTION, DIA=DIALOGUE)

puts: mean 0.51, standard deviation 0.14, indicating reasonable causal consistency with differentiation enabling RL optimization.

Limitation Analysis. Zero-shot approach exhibits domain gaps with fantasy-specific causality. Example: "Cast Fireball" $\rightarrow$ "Goblins scatter" scored 0.89 in one test but similar fantasy cause-effect pairs occasionally score lower due to NLI training on non-fantasy corpora. Despite this limitation, the critic provides usable reward signals distinguishing appropriate versus inappropriate responses.

6.5 Hybrid Player System Development

Dual-Component Architecture Implementation.

Generator Component: Fine-tuned DistilGPT-2 (82M parameters) on 519,597 player utterances from combined CRD3+LIGHT corpus. Model generates diverse player prompts across exploration, combat, and social interaction scenarios during RL training.

Classifier Component: Trained DistilBERT-base (66M parameters) for three-way intent classification with keyword-based automatic labeling. Categories: EXPLORE (investigation, movement), ACTION (combat, skills), DIALOGUE (conversation, roleplay).

Integration Results. Table 7 demonstrates generation and classification performance.

Quality Category	Mean	Below Thr.
Narrative Quality	0.38 ± 0.09	57%
Causal Consistency	0.51 ± 0.14	18%

Table 8: Baseline quality assessment ($N = 100$)

Classification Performance. Mean confidence across categories: EXPLORE 0.87, ACTION 0.91, DIALOGUE 0.84. High confidence enables reliable dynamic weight selection during RL training. Intent distribution: EXPLORE 35%, ACTION 40%, DIALOGUE 25%.

6.6 Analytical Findings

Finding 1: Supervised Learning Limitations. Baseline model achieves grammatical coherence but exhibits systematic quality weaknesses quantified through critic evaluation:

Over half of baseline outputs score below acceptable narrative threshold (0.4), empirically demonstrating supervised training's insufficiency for quality generation despite 310K training examples. This validates the necessity of critic-guided optimization.

Finding 2: Critic Evaluation Validity. Both critics provide differentiated, meaningful quality signals:

Narrative critic achieves Pearson correlation 0.938 on validation data, successfully ranking quality from rich prose (0.64-0.72) to generic text (0.28-0.29). Score distribution on baseline outputs ($\mu = 0.38$, $\sigma = 0.09$) shows variance enabling fine-grained discrimination despite overall low quality.

Causal critic distinguishes logical responses (0.78-0.89) from non-sequiturs (0.12-0.15). Distribution on baseline ($\mu = 0.51$, $\sigma = 0.14$) indicates reasonable consistency with room for targeted improvement.

Both critics demonstrate score ranges substantially exceeding noise levels, validating their utility as RL reward signals.

Finding 3: Multi-Objective Trade-off Evidence. Correlation analysis between narrative and causal scores on baseline outputs reveals weak negative correlation ($\rho = -0.18$, $p = 0.07$). Subset analysis shows responses scoring high causal (¿0.6, $N = 21$) average narrative 0.36 versus mean 0.38 overall, while high narrative responses (¿0.45, $N = 9$) average causal 0.44 versus mean 0.51.

This preliminary evidence suggests inherent tension between objectives: elaborative narrative description may compromise causal precision, while strict logical consistency may limit atmospheric elaboration. This trade-off directly motivates multi-objective optimization with context-dependent weighting—different scenarios should prioritize different objectives rather than forcing uniform balance.

Finding 4: Dynamic Weighting Justification. Intent distribution analysis reveals: EXPLORE 35%, ACTION 40%, DIALOGUE 25%. These categories exhibit different quality requirements: exploration benefits from atmospheric description (high narrative priority), action demands consequence precision (high causal priority), dialogue requires balanced quality. This heterogeneity validates dynamic weighting design over static uniform weights.

6.7 PPO Framework Development

Implementation of complete PPO training infrastructure with multi-critic integration and dynamic weighting mechanisms. Framework components: policy model (initialized from SFT baseline), reference model (frozen baseline for KL penalty), multi-critic reward computation, intent-based weight selection, and PPO trainer with stability mechanisms (KL penalty, reward normalization, clipping).

Training pipeline validated through initial runs; full-scale multi-variant training in progress for final submission.

7 Timeline

7.1 Completed (Weeks 1-4)

Week 1: Data Acquisition and Preprocessing

- Downloaded CRD3 (324 episodes, 611K utterances) and LIGHT (108K responses)
- Implemented parsing pipelines with context window management
- Created unified 310K corpus with stratified splits

Week 2: Baseline Development

- Integrated datasets with standardized instruction formatting
- Trained SFT baseline (Phi-2 + QLoRA on 26K subset, 2.8 hours)
- Baseline evaluation: narrative 0.38, causal 0.51 mean scores

Week 3: Narrative Critic Development

- Constructed 41K critic dataset (30K ROCStories + 11K D&D + 30K synthetic)
- Trained DeBERTa regression model (Pearson 0.938, MAE 0.32)
- Validated discrimination capability (0.28-0.72 score range)

Week 4: Causal Critic and Player System

- Implemented zero-shot causal critic (DeBERTa-NLI entailment scoring)
- Trained hybrid player generator (DistilGPT-2 on 520K utterances)
- Trained intent classifier (DistilBERT, 0.82+ confidence)
- Built PPO training infrastructure with multi-GPU support

7.2 In Progress (Week 5)

- Developing world consistency critic (symbolic tracker + Flan-T5 extractor)
- Developing character voice critic (DeBERTa on CRD3 NPC sequences)
- Training RL-Narrative and RL-Causal single-critic variants
- Initial RL-Combined experiments with static weights

7.3 Planned (Weeks 6-8)

Week 6: Multi-Critic Integration and Training

- Complete world consistency and character voice critics
- Finalize RL-Combined variant with all four critics
- Train full Director MCRL with dynamic intent-based weighting
- Generate test outputs from all five model variants

Week 7: Comprehensive Evaluation

- Intrinsic metrics: four critic scores, perplexity, diversity
- Intent-stratified analysis (EXPLORE/ACTION/DIALOGUE)
- Extrinsic evaluation: Story Cloze, HellaSwag, consistency probes
- Statistical testing and ablation studies

Week 8: Analysis and Finalization

- Qualitative analysis: comparative outputs, failure categorization
- Results visualization and trade-off analysis
- Final report compilation with complete methodology

- Code documentation and submission preparation

7.4 Key Milestones

Milestone	Deadline	Status
Core critics validated	Week 4	Complete
All four critics complete	Week 6	In Progress
RL variants trained	Week 6	Planned
Evaluation complete	Week 7	Planned
Final submission	Week 8	Planned

Table 9: Project milestones and completion status

8 Discussion

8.1 Methodological Validation

Our interim results support the MCRL approach through three key findings:

1. Baseline Limitations: Mean narrative score 0.38 with 57% below threshold demonstrates supervised training's insufficient for quality generation, empirically motivating critic-guided optimization.

2. Critic Differentiation: Substantial score variance (narrative 0.28-0.72, causal 0.12-0.89) indicates critics capture meaningful quality dimensions enabling targeted RL improvement.

3. Objective Tension: Preliminary correlation analysis suggests inverse relationship between narrative elaboration and causal precision, validating multi-objective framework necessity.

8.2 Limitations

Model Scale: Phi-2 (2.7B) versus proposed Llama-2-7B (7B) reduces absolute generation quality. However, methodology remains model-agnostic and transfers to larger models.

Critic Domain Coverage: Zero-shot causal critic shows fantasy-domain limitations. Fine-tuning on D&D-specific pairs would improve coverage.

9 Broader Impact

The MCRL framework provides a general methodology for domains requiring principled multi-constraint satisfaction beyond entertainment applications.

Trustworthy AI Systems. Specialized critics for factual consistency, logical soundness, and citation quality enable verifiable research assistants. Multi-objective optimization prevents trade-offs where factual accuracy compromises readability, or comprehensiveness sacrifices precision.

Professional Domain Assistants. Legal systems require precedent consistency and document coherence; medical applications demand clinical guideline adherence and HIPAA compliance; software engineering needs correctness, security, and readability. Dynamic weighting adapts critic priorities to task context—exploratory legal research prioritizes comprehensiveness while contract drafting emphasizes precision.

Safety-Critical Applications. High-stakes domains benefit from explicit decomposition into verifiable objectives. Rather than monolithic "safety" scores, systems can separately optimize correctness, interpretability, and risk mitigation with transparent weighting policies.

Limitations. Critic quality bounds overall system performance—poorly calibrated critics propagate errors through RL training. Dynamic weighting requires accurate intent classification; misclassification applies inappropriate objective priorities. The framework assumes objectives can be meaningfully decomposed and independently evaluated, which may not hold for deeply entangled constraints.

10 Conclusion

Interactive narrative generation exemplifies a fundamental challenge in language modeling: balancing competing objectives that often exist in tension. Standard supervised approaches, while achieving surface-level fluency, systematically fail to capture the multi-dimensional quality requirements of complex generation tasks. Our empirical analysis demonstrates this limitation—57% of supervised baseline outputs score below quality thresholds despite training on 310K examples, exhibiting generic vocabulary, repetitive structures, and vague consequences.

We address this through Multi-Critic Reinforcement Learning with dynamic reward weighting, decomposing generation quality into specialized, measurable competencies. The Director LLM framework trains dedicated critics for narrative quality and causal consistency, achieving strong discriminative capability (narrative Pearson 0.938; causal zero-shot discrimination 0.78-0.89 vs. 0.12-0.15). Dynamic weighting based on classified intent enables context-appropriate objective prioritization—exploration scenarios emphasize atmospheric description while combat demands consequence precision.

Our approach contributes three key insights. First, complex generation tasks benefit from explicit objective decomposition rather than monolithic reward modeling. Second, competing objectives require adaptive prioritization; static uniform weights sub-optimally balance heterogeneous contexts. Third, specialized critics providing differentiated signals enable targeted optimization through reinforcement learning where supervised methods plateau.

The framework generalizes beyond interactive storytelling to any domain requiring multi-constraint satisfaction with context-dependent priorities: trustworthy AI systems balancing accuracy and accessibility, professional assistants navigating domain compliance and user preferences, safety-critical applications optimizing correctness and interpretability. By treating multi-objective generation as a principled decomposition and adaptive weighting problem, MCRL provides a path toward more controllable, coherent, and capable language models.

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